k0n53 walked descent: walks per shared inversion

Date: 2026-11-04. **Repository:** `aburan28/cryptanalysis`, production Rust crate `suite/`; study `experiments/descent-walks-k0n53-20261104/`. **Protocol:** `PROTOCOL.md`, frozen in commit 20228687b before any data. **Class:** engineering. The verified logarithms are unchanged; the order of probes changes. **Isolation:** L0 (shared 4-vCPU VM, CPU 3 pinned, single thread).

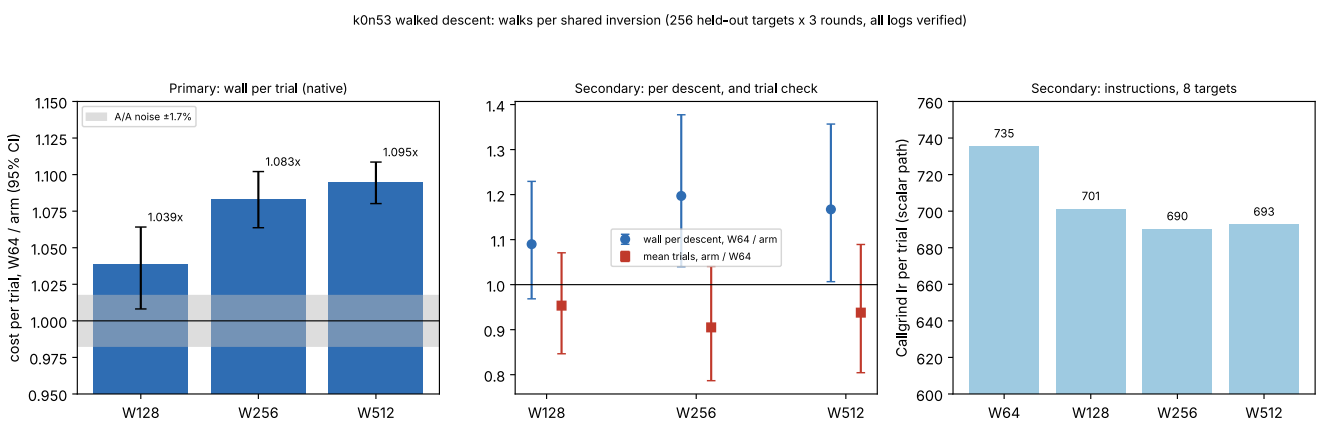


Figure 1: Cost per trial against 64 walks (paired target bootstrap, A/A band); per-descent wall and the mean-trials check; Callgrind Ir per trial.

Result

All 3,840 descents (256 held-out targets × 5 arms × 3 rounds) solved, and every logarithm equals its target's scalar [pcc:unverified].

Cost per trial against 64 walks:

- W128: 1.039× (95% CI 1.008–1.064);
- W256: 1.083× (1.064–1.102);
- **W512: 1.095× (1.080–1.109)**.

The A/A pair is 0.992× (0.975–1.009), a noise floor of ±1.7%. Native cost falls from 96.6 to 88.2 ns per trial, and Callgrind Ir per trial from 735 to 693 [pcc:unverified]. Mean trials per descent stay consistent with 64 walks: W512 is 0.938× (0.804–1.089), as predicted for uniform probes. **Decision (protocol rule): adopt W512.**

Scope of the adoption

The protocol's rule names the global default. After the run, a perfbench check found that 512 walks on every curve made `relation/koblitz_rung_k0n31` ($r \approx 2^{29}$) 0.835× as fast: a descent there takes too few trials to repay starting 512 walks. The protocol measured only k0n53 ($r \approx 2^{44.3}$). The change therefore sets 512 walks only on subgroups of order at least 2^{40}; smaller ones keep 64. This deviation is reported here, and the sizes between the two measured points are not measured. Against `main`, the scoped change makes the perfbench k0n53 descent 1.222× faster (1.088–1.362; its fingerprint changes, as predicted) [pcc:unverified]. The k0n31 rung, collection and signed rho are neutral.

Limits

This is a descent-phase gain on one curve and one host. It is not a complete-solve S, and it makes no matched-rho claim.